

1. What is the average GRE score of an applicant who has been accepted into a PhD program?

The average GRE score of applicants who were accepted into PhD programs in the dataset was 251.42. This value was calculated by filtering the dataset to include only rows where the degree type was PhD and the application outcome was recorded as Accepted, and then computing the mean GRE score for those entries. The result provides a general indication of the standardized test performance of successful PhD applicants in the dataset. However, the value should be interpreted cautiously because the Grad Café dataset is based on self-reported information, which may include inconsistencies or inaccuracies in test score reporting.

2. What is the average GPA of an applicant who has been accepted into a Master's program?

The average GPA of applicants who were accepted into Master's programs was 3.69 on a 4.0 scale. This statistic was computed by filtering the dataset to include only rows where the degree type was Master's and the outcome was Accepted, and then calculating the mean GPA across those records. The result suggests that successful Master's applicants generally have strong undergraduate academic performance. However, the value may still be influenced by self-reported entries and variations in grading systems across different institutions.

3. What are 2-3 shortcomings of inferring decision\_date from a text status field and the term value?

There are several limitations when inferring the decision date from the status text and the application term. First, the status field may contain inconsistent formatting, making it difficult to reliably extract dates using automated parsing methods. Second, inferring the year from the application term may produce incorrect results if decisions occur in a different calendar year than expected. Third, some records may not include an explicit decision date at all, which results in missing or partially inferred values that reduce the accuracy of the derived decision\_date column.

4. What did your correlation analysis suggest about the relationship between GPA and GRE-related measures?

The correlation analysis examined the relationships between GRE total scores, GRE verbal scores, and undergraduate GPA. The Pearson correlation between GRE total score and GRE verbal score was 0.242, with a very small p-value ( $7.56 \times 10^{-25}$ ), indicating a statistically significant positive relationship between these two variables. This result is expected because the verbal section contributes directly to the overall GRE score. In contrast, the correlation between GPA and GRE score was  $-0.036$ , with a p-value of 0.107, suggesting that there is essentially no meaningful relationship between GPA and GRE performance in this dataset. Because the p-value is greater than 0.05, the

relationship between GPA and GRE scores is not statistically significant, indicating that undergraduate GPA and standardized test performance may reflect different aspects of academic ability.

5. Did your group comparison suggest that accepted and rejected applicants differ in GPA? Briefly explain.

A two-sample t-test was conducted to compare GPA values between applicants who were accepted and those who were rejected. The analysis included 6,401 accepted applicants and 6,329 rejected applicants, using only GPA values between 0 and 4. The test produced a statistic of  $-4.64$  with a p-value of  $3.49 \times 10^{-6}$ , indicating that the difference in GPA between the two groups is statistically significant. Because the p-value is far below the conventional threshold of 0.05, we can conclude that accepted and rejected applicants differ in GPA on average. However, while the difference is statistically significant due to the large sample size, GPA alone is unlikely to fully determine admission outcomes, as admissions decisions also depend on other factors such as research experience, recommendation letters, and program competitiveness.

6. What did your chi-square test suggest about the relationship between your selected categorical variables?

A chi-square test was conducted to evaluate whether there is an association between degree type and application outcome. The contingency table compared the number of accepted, interviewed, rejected, and waitlisted applicants across Master's and PhD degree categories. The test produced a chi-square statistic of 4335.34 with 3 degrees of freedom and a p-value effectively equal to 0, indicating a highly statistically significant association between degree type and application outcome. Because the p-value is far below the conventional significance threshold of 0.05, we reject the null hypothesis that degree type and outcome are independent. This result suggests that the distribution of application outcomes differs between Master's and PhD applicants within the dataset.

7. What data quality issues still remain in the cleaned dataset, even after your processing?

Despite the cleaning and validation steps applied during this analysis, several data quality issues still remain in the dataset. First, the data originates from self-reported entries on Grad Café, which means that some values may be inaccurate, incomplete, or inconsistently reported by users. Second, certain numeric fields such as GPA and GRE scores may still contain unrealistic or extreme values that reflect data entry errors or different grading systems used internationally. Third, some columns continue to contain missing values, particularly for GRE subscores, GPA, and decision dates, which can limit the completeness of statistical analyses. Finally, text-based fields such as program names and universities may still contain variations in spelling or formatting that make it difficult to consistently group or aggregate results. These remaining issues highlight the

limitations of working with crowdsourced datasets and suggest that additional validation or standardization would be beneficial for future analyses.